



Introduction

The exponential proliferation of electric vehicles, consumer electronics, and portable industrial devices has globally generated a massive volume of decommissioned, discarded lithium-ion batteries. While these cells are often retired from their primary high-demand applications, many of them still retain substantial valuable capacity suitable for less-demanding second-life applications such as stationary energy storage systems or secondary consumer goods. However, traditional industry methods for cell grading rely on lengthy charge-discharge cycles or prohibitively expensive laboratory machinery, rendering large-scale screening incredibly slow and commercially impractical. To address this severe operational bottleneck, we developed **Smart Sort**: a highly portable, low-cost battery health analyser that accurately evaluates a cell's true State of Health (SoH) in just a matter of a few seconds by analysing rapid transient behaviours.

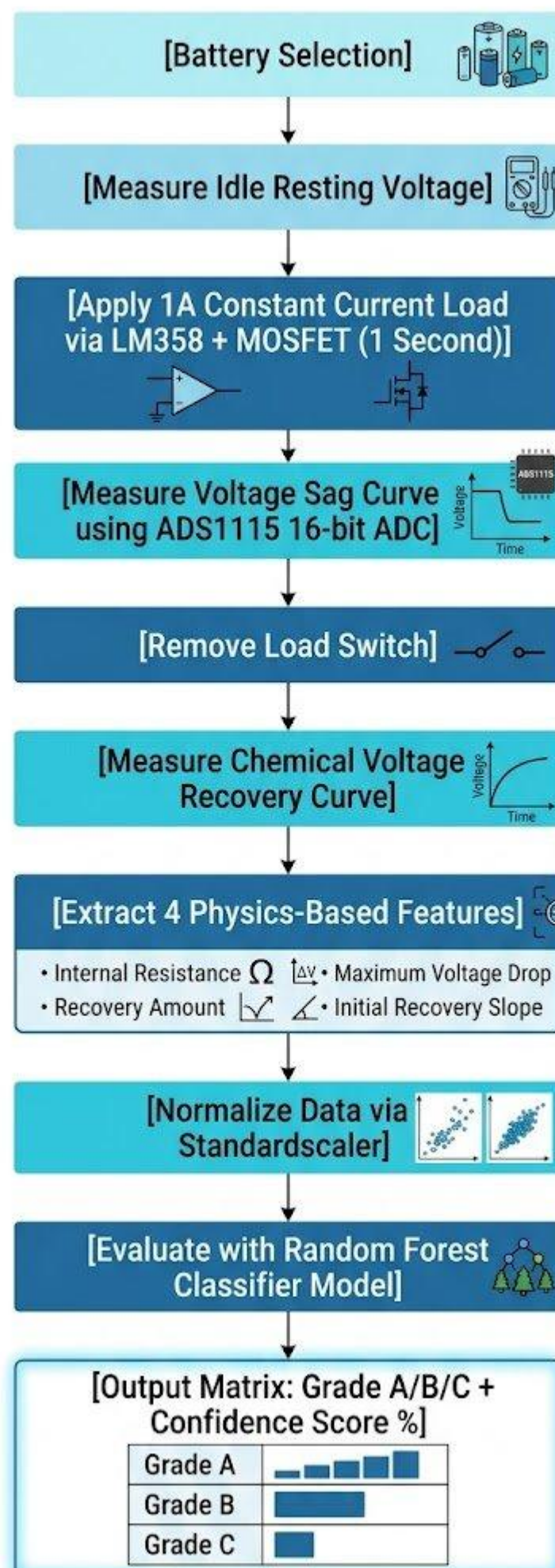
Problem Statement

The lack of a rapid, scalable, and decentralized infrastructure to grade "loose" retired lithium-ion cells directly prevents their widespread reuse in secondary markets. This severe technological gap forces completely viable batteries into premature, hazardous recycling processes or, worse, drives them directly into landfills, creating a compounding electronic waste crisis and wasting substantial residual energy capacity.

Objectives

- Develop a portable system for rapid battery health assessment of lithium-ion cells.
- Estimate the true State of Health (SoH) within seconds using dynamic load-response analysis.
- Replace slow, multi-hour charge-discharge testing with a fast, low-cost, and reliable solution.

Methodology



Tools used

Hardware

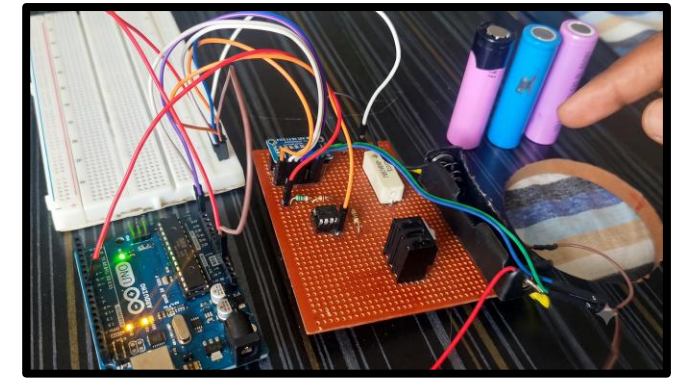
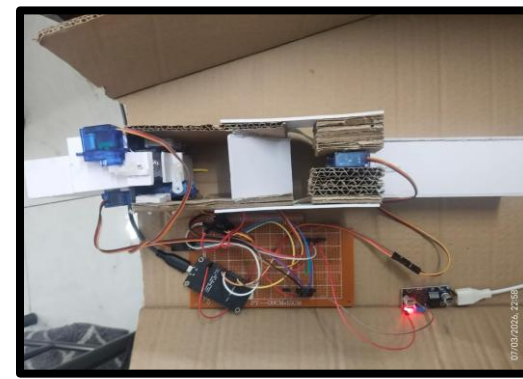
Arduino Uno & ESP32
ADS1115 (16-bit ADC), LM358 Op-Amp, IRF540N MOSFET
1 Ω Shunt Resistor, DC-DC Boost Converter
SG90 Servo Motors ($\times 4$), Breadboard & Jumper Wires

Software

Python 3, Arduino IDE
LTspice, Scikit-learn (StandardScaler & RandomForest)

Results and Discussions

- Discrete Cell Classification:** The Smart Sort system successfully classified retired lithium-ion cells into discrete Grade A, Grade B, and Grade C categories by analysing their transient load-and-recovery curves under specific current pulses.
- Feature Extraction Matrix:** Key extracted engineering features—such as instantaneous internal resistance drops, dynamic voltage sag under load, and subsequent chemical recovery profiles—enabled highly accurate estimation of each cell's true State of Health (SoH).
- Time Framework Compression:** The fully functional prototype successfully compresses overall testing validation time frames from several hours down to a few seconds, providing an exceptionally low-cost and scalable screening method.



Measurable Outcomes

- Functional Bottleneck Solution:** The Smart Sort prototype successfully demonstrates a functional, low-cost solution to the massive testing bottleneck currently hindering second-life lithium-ion battery markets.
- High-Resolution Transient Analysis:** By replacing hours of traditional charge-discharge cycling with rapid, high-resolution transient load analysis, this embedded system establishes that a cell's State of Health can be determined accurately within seconds.
- Industrial Scalability & Sustainability:** Implementing this automated sorting technology at scale will play a critical role in optimizing industrial battery recycling pipelines, reducing electronic waste, and making sustainable energy storage far more accessible.

References

- Natella, D., Onori, S., & Vasca, F. (2022). A co-estimation framework for state of charge and parameters of lithium-ion battery with robustness to aging and usage conditions. *IEEE Transactions on Industrial Electronics*, 69(12), 12489–12499. <https://ieeexplore.ieee.org/document/9856595/authors#authors>
- Lai, X., Yuan, M., Tang, X., Yao, Y., Weng, J., Gao, F., Ma, W., & Zheng, Y. (2021). Co-estimation of state of charge and state of health for lithium-ion batteries using an enhanced electrochemical model. *IEEE Transactions on Power Electronics*, 36(6), 6302–6313. <https://ieeexplore.ieee.org/document/9384220>
- Heeger, D., Partridge, M., Trullinger, V., & Wesolowski, D. (2017). *Lithium battery health and capacity: Estimation techniques using embedded electronics* (SAND2017-10722). Sandia National Laboratories. <https://www.osti.gov/servlets/purl/1596204>

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